

| Question |  |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks available |     |     |       |       |      |
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|          |  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 7        |  |  |  | <p><b>Indicative content</b></p> <p><b>Advantages</b><br/>good for healthy bones and teeth (due to calcium ions)<br/>reduces risk of heart disease<br/>good for brewing beer<br/>better taste</p> <p><b>Disadvantages</b><br/>does not readily form lather with soap solution / forms scum<br/>more soap solution needed to produce lather for washing e.g. clothes, dishes<br/>boilers and hot water pipes become ‘furred up’ (due to calcium carbonate / limescale precipitating)<br/>boilers and kettles become less efficient and pipes and radiators can become completely blocked<br/>energy is wasted as a result</p> <p><b>5-6 marks</b><br/>Detailed description of advantages and disadvantages of hard water; two examples of each with reference to consequences of disadvantages<br/><i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i></p> <p><b>3-4 marks</b><br/>Basic description of advantages and disadvantages of hard water; one example of each with simple development of one point<br/><i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i></p> <p><b>1-2 marks</b><br/>Simple reference to one advantage and one disadvantage<br/><i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i></p> <p><b>0 marks</b><br/><i>No attempt made or no response worthy of credit.</i></p> | 6               |     |     | 6     |       |      |
|          |  |  |  | Question 7 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 6               | 0   | 0   | 6     | 0     | 0    |